UNIT 1 · STANDARD 6.NS.2

Divide Multi-Digit Numbers

Lesson 1-4

Name: ____________ Date: ____________ Class: ____________

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

In 156 ÷ 12 = 13, the dividend is 156 — it is the total being split up

Dividend

The number you are splitting up in a division problem.

In 156 ÷ 12 = 13, the divisor is 12 — it is the size of each group

Divisor

The number you split by in a division problem.

In 156 ÷ 12 = 13, the quotient is 13 — there are 13 groups

Quotient

The answer when you divide.

17 ÷ 5 = 3 R 2 means 3 groups of 5 with 2 left over

Remainder

What is left over when a number does not divide evenly.

1,344 ÷ 12: first 12 × 100 = 1,200, then 12 × 12 = 144. Quotient = 100 + 12 = 112

Partial quotients

A way to divide by breaking the problem into smaller, easier steps.

Key Ideas & Notes

🎯 **WHAT WE'RE LEARNING TODAY**

I can divide multi-digit whole numbers and explain the meaning of the quotient and remainder.

1️⃣ Notice

The space station has 1,344 nutrition bars that must be divided equally among 12 crew sections for a 4-week rotation.

2️⃣ Key idea

I can divide multi-digit whole numbers and explain the meaning of the quotient and remainder.

3️⃣ Words I'll use

Dividend **Divisor** **Quotient** **Remainder** **Partial quotients**

🤔 Think About It

- What total amount needs to be divided?
- How many groups are the supplies being split into?
- Will each section get the same amount with none left over?

See the notes in action: watch one worked all the way through, then try the next with the same steps.

👀 I do — watch

Follow each step as your teacher solves it.

Problem: What is 936 ÷ 12?

A. 78
B. 87
C. 68
D. 76

Step 1 $12 \times 78 = 936$.

Step 2 You can verify: $12 \times 70 = 840$, $12 \times 8 = 96$, $840 + 96 = 936$.

✅ **Answer:** A. 78

🤝 **We do — together**

Solve this one with your class using the same steps.

Problem: What is 2,485 ÷ 5?

A. 497

B. 487

C. 507

D. 496

Step 1 ______________________

Step 2 ______________________

Answer: ______________________

✏️ **You do — your turn**

Now try one on your own. Show every step.

Problem: What is 756 ÷ 9?

A. 84

B. 83

C. 94

D. 86

Show your work:

____

____

____

Turn & Talk — Discussion Points

Level 1 support | Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

The station must divide 1,344 nutrition bars equally among 12 crew sections. Before dividing, about how many bars should each section get, and how do you know?

Level 1 support

Start here: Estimating with friendly numbers checks whether the quotient is reasonable.

💡 **Need a hint?**

Sentence starters (optional)

Each section gets about ____ bars because 1,344 is close to ____.

Cada sección recibe unas ____ barras porque 1,344 está cerca de ____.

I estimated by ____.

Estimé al ____.

Word bank: dividend | divisor | quotient | estimate | partial quotients

Listen for: *Students estimate using 1,200 ÷ 12 = 100 (or similar) to predict roughly 100+ bars per section, identifying 1,344 as the dividend and 12 as the divisor.*

Level 2

Will 1,344 divide evenly by 12, or will there be a remainder? Predict before computing and justify your reasoning.

- I predict the remainder is ____ because ____.
- 1,344 is divisible by 12 since ____.

EXPLORE

You used partial quotients for 1,344 ÷ 12. What steps did you use, and how do they build to the final quotient?

Level 1 support

Start here: Partial quotients break a hard division into easy chunks you add together.

💡 **Need a hint?**

Sentence starters (optional)

First I took 12 × ___ = 1,200, then 12 × ___ = 144.

Primero usé 12 × ___ = 1,200, luego 12 × ___ = 144.

The quotient is ___ because I added the partial quotients ___.

El cociente es ___ porque sumé los cocientes parciales ___.

Word bank: partial quotients | dividend | divisor | quotient | remainder

Listen for: *Students explain 12 × 100 = 1,200, then 144 left, 12 × 12 = 144, so the quotient is 100 + 12 = 112 with no remainder.*

Level 2

Why does it not matter whether you start with 12 × 100 or 12 × 50 in partial quotients? Will you reach the same quotient?

- It does not matter where I start because ___.
- Different first chunks still give 112 since ___.

CONNECT

A school divides 1,680 pencils equally among 24 classrooms. How does multi-digit division tell each classroom's share, and how can you check it?

Level 1 support

Start here: Dividing 1,680 by 24 gives each classroom's count; multiply back to check.

💡 **Need a hint?**

Sentence starters (optional)

Each classroom gets ____ pencils because 1,680 ÷ 24 = ____.

Cada salón recibe ____ lápices porque 1,680 ÷ 24 = ____.

I can check by multiplying ____ × ____ = 1,680.

Puedo comprobar multiplicando ____ × ____ = 1,680.

Word bank: **dividend** **divisor** **quotient** **partial quotients** **equally**

Listen for: *Students compute 1,680 ÷ 24 = 70 and verify with 24 × 70 = 1,680, explaining the quotient is the per-classroom amount.*

Level 2

If the school had 1,700 pencils instead, what would the quotient and remainder be, and what would the remainder mean in this situation?

- With 1,700 pencils the quotient is ____ and the remainder is ____.
- The remainder means ____ in real life because ____.

REFLECT

In a division problem, what is the difference between the quotient and the remainder, and what does each one tell you?

Level 1 support

Start here: The quotient is the size of each equal share; the remainder is what is left over.

💡 **Need a hint?**

Sentence starters (optional)

The quotient tells me ____, and the remainder tells me ____.

El cociente me dice ____, y el residuo me dice ____.

A remainder of 0 means ____.

Un residuo de 0 significa ____.

Word bank: **quotient** **remainder** **dividend** **divisor** **divide**

Listen for: *Students explain the quotient is the number in each group while the remainder is the amount left that cannot be shared equally, and a remainder of 0 means it divides evenly.*

Level 2

When is it correct to round a quotient UP because of a remainder, and when should you drop the remainder? Give an example of each.

- I round up when ____, for example ____.
- I drop the remainder when ____, for example ____.

Write About the Math

The Writing Revolution

I can explain my division using the words dividend, divisor, quotient, and remainder.

1. Kernel Sentence

subject + verb

Model: Partial quotients is a way to divide by breaking the problem into smaller, easier steps.
Cocientes parciales es una manera de dividir separando el problema en pasos más fáciles.

Write a kernel sentence about partial quotients. Use a subject and a verb.
Escribe una oración base sobre cocientes parciales. Usa un sujeto y un verbo.

___

___

2. Sentence Expansion

because · but · so

Kernel: Partial quotients matters in math
Cocientes parciales importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because *porque*

Partial quotients matters in math because ____.
Cocientes parciales importa en matemáticas porque ____.

___

but *pero*

Partial quotients matters in math, but ____.
Cocientes parciales importa en matemáticas, pero ____.

___

so *entonces*

Partial quotients matters in math, so ____.
Cocientes parciales importa en matemáticas, entonces ____.

___

3. Sentence Types

4 ways to write a math idea

Statement
Afirmación

Tell one true fact about partial quotients.
Di un hecho verdadero sobre partial quotients.

Partial quotients ____.

Question
Pregunta

Ask a question about partial quotients.
Haz una pregunta sobre partial quotients.

How does ____ ?
¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about partial quotients.
Muestra entusiasmo sobre partial quotients.

Wow, ____ !
¡Guau, ____ !

Command
Mandato

Tell a partner what to do with partial quotients.
Dile a un compañero qué hacer con partial quotients.

First, ____ .
Primero, ____ .

4. Explain Your Reasoning

use a sentence starter

Each section gets about ____ bars because 1,344 is close to ____.
Cada sección recibe unas ____ barras porque 1,344 está cerca de ____.

I estimated by ____.
Estimé al ____.

First I took 12 × ____ = 1,200, then 12 × ____ = 144.
Primero usé 12 × ____ = 1,200, luego 12 × ____ = 144.

Try It

Solve on your own. Check the answer key when you are done.

1. A school orders 3,648 pencils packed 24 to a box. How many boxes are there?

A. 152

B. 148

C. 160

D. 144

Show your work:

2. What is 5,084 ÷ 31?

A. 164

B. 154

C. 174

D. 164 r2

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A school has 1,680 pencils to distribute equally to 24 classrooms. Show how to solve this using partial quotients. Explain each step and check your answer by multiplying.

Sentence starter: 1,680 ÷ 24 = ____. First I used 24 × ____ = ____, then 24 × ____ = ____. The partial quotients add to ____. I checked by multiplying: 24 × ____ = 1,680.

Show your work:

Reflect — Exit Ticket

What is 2,352 ÷ 16?

A. 147

B. 137

C. 157

D. 146

Your answer:

Answer Key & Teacher Guide

1. **We Try (notes):** A. 497 *— 5 × 497 = 2,485. Check: 5 × 400 = 2,000, 5 × 90 = 450, 5 × 7 = 35. 2,000 + 450 + 35 = 2,485.*
2. **You Try (notes):** A. 84 *— 9 × 84 = 756. Check: 9 × 80 = 720, 9 × 4 = 36, 720 + 36 = 756.*
3. **Try It 1:** A. 152 *— 3,648 ÷ 24 = 152 boxes.*
4. **Try It 2:** A. 164 *— 31 × 164 = 5,084 exactly, so there is no remainder.*
5. **Exit Ticket:** A. 147 *— 16 × 147 = 2,352. Check: 16 × 100 = 1,600, 16 × 40 = 640, 16 × 7 = 112. 1,600 + 640 + 112 = 2,352.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Partial quotients is a way to divide by breaking the problem into smaller, easier steps.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).